Light Speed Calculations

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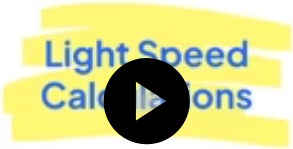

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Light speed calculations

- Light is a type of electromagnetic wave which travels at a constant speed of 3×10^8 m/s
- Objects in the Solar System are visible from Earth as they **reflect light** from the Sun
- It takes time for light to travel such large distances, for example:
  - it takes **8 minutes** for light from the Sun to reach Earth
  - it takes around **5 hours** for light from the Sun to reach the outer regions of the Solar System
  - it takes **4 years** for light from our nearest star (after the Sun) to reach Earth
- The Milky Way galaxy contains billions of stars, huge distances away, with the light taking even longer to be seen from Earth
- To carry out light speed calculations, we can rearrange the equation:

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}$$

- So, the time taken for light to travel a distance can be calculated using:

$$\text{time} = \frac{\text{distance}}{\text{speed of light}}$$

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Worked Example

The radius of Mercury's orbit around the Sun is 5.8×10^{10} m.

The speed of light is 3.0×10^{8} m/s.

Calculate the time taken for light from the Sun to reach Mercury.

Answer:

Step 1: List the known quantities

- Distance travelled = 5.8×10^{10} m
- Speed of light = 3.0×10^{8} m/s

Step 2: State the equation for the time taken for light to travel a distance

$$\text{time} = \frac{\text{distance}}{\text{speed of light}}$$

Step 3: Substitute the values into the equation

$$\text{time} = \frac{5.8 \times 10^{10}}{3.0 \times 10^{8}} = 193 \text{ s}$$

- It takes about 193 s, or 3 min, for light from the Sun to reach Mercury

Examiner Tips and Tricks

The speed of light is **very** fast. This is why in our everyday life things like switching on a light seem to be instant. However, this is only because the light travels very fast and the distances are very small. In large, astronomical distances which can be millions or even billions of kilometres, the limit of the speed of light starts to have an effect.

For example, it takes light 8 minutes to travel from the Sun to the Earth. This means we are seeing the Sun as it was eight minutes ago. If the Sun was to disappear, we would not notice till eight minutes later. Although, by that time, time delay would be the least of our worries...

p.s.: The Sun is not going to vanish!

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